

EECS 496 – MDE Professionalism Introduction

Instructors

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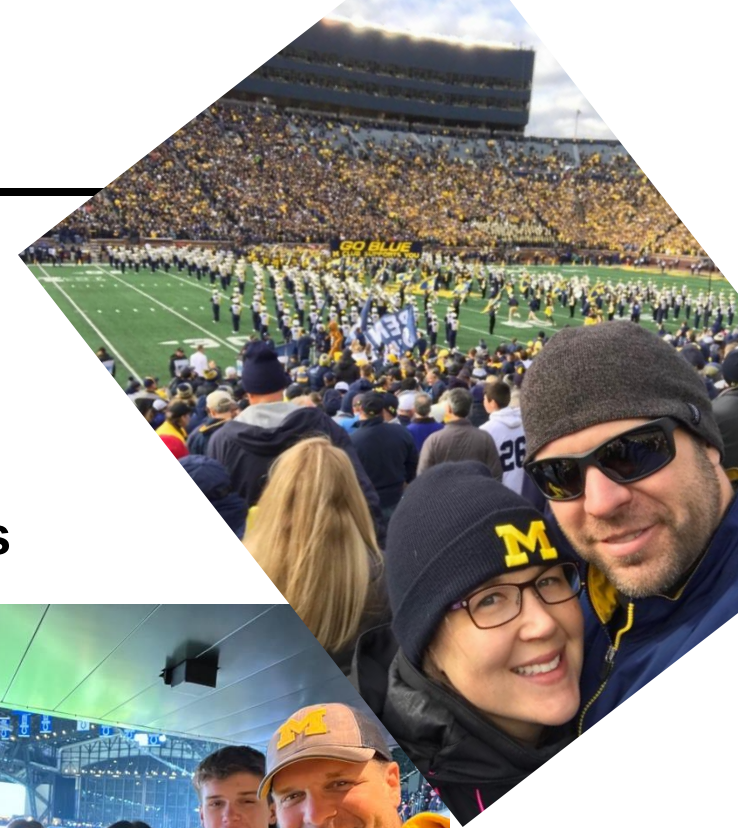
GSI: Lexie Wang, lexzww@umich.edu

IA: Maatla Sefawe, maatla@umich.edu



Prof. David Wentzloff

- **Background**
 - Grew up in Beulah, MI (near Traverse City)
 - Frankfort High School – graduating class of 31 students
 - Then: Football, basketball, and track
 - Now: watersports, skiing, home automation and IoT gadgets
- **Education**
 - **UM BSE**, MIT MS and PhD
- **Teaching experience (mainly circuits)**
 - ENG 100, EECS 215, 311, 413, 496, 522
- **Research experience**
 - Low power, radio frequency (RF) integrated circuits
 - Batteryless sensors and energy-harvesting
- **Industry Experience**
 - Ford and Intel
 - Consulting at small and big companies



Entrepreneurial Experience



Founder, BoD, CTO
A2 office (& VA, CA)



Founder, BoD
A2 office (& CA, Toronto)



Founder
A2 office

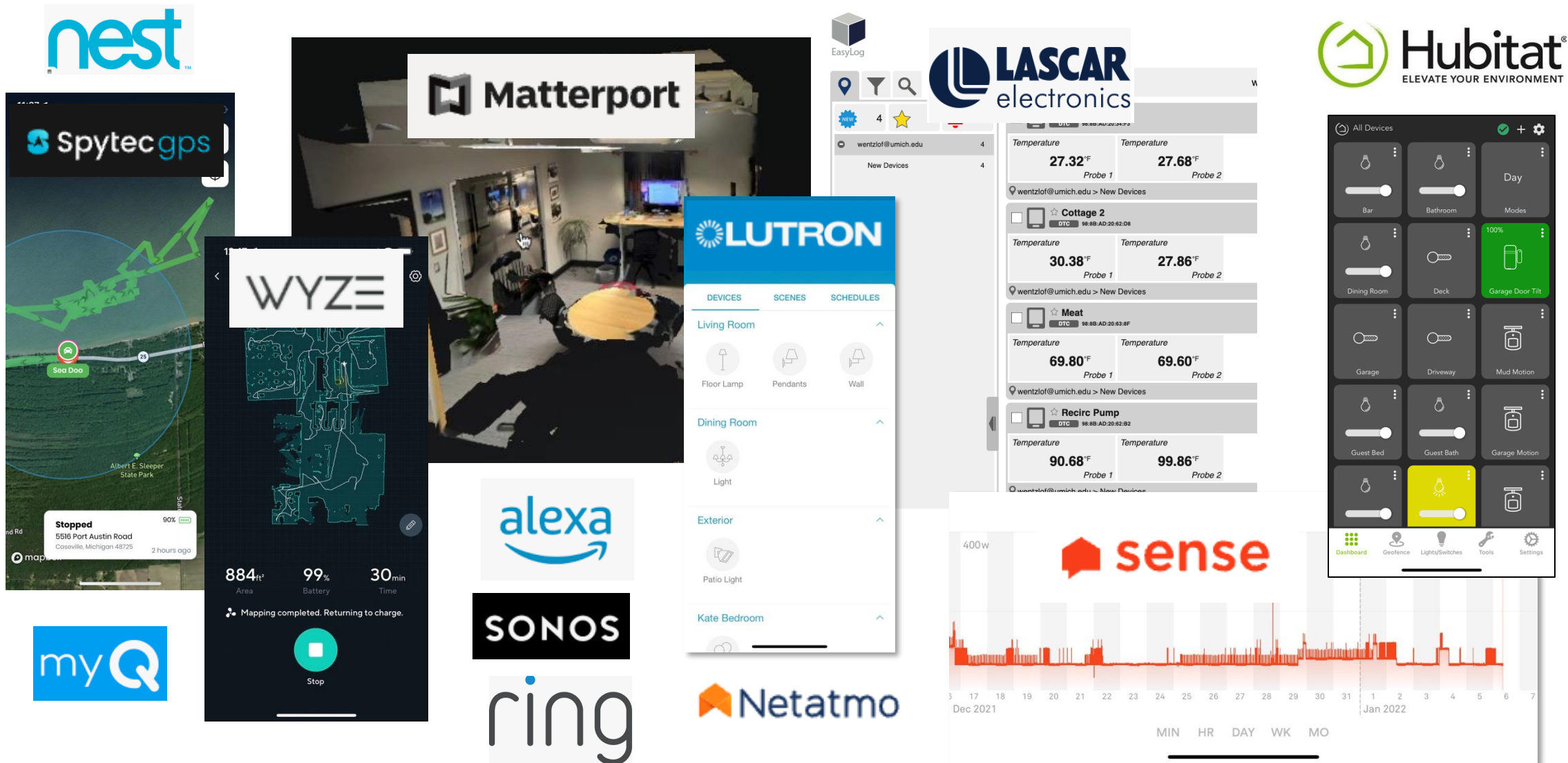


Adviser
CA office



Former Faculty Director

My Interests – IoT



Team and Communication

- GSI: Lexie Wang (lexzww@umich.edu)
- IA: Maatla Sefawe (maatla@umich.edu)
- Staff email address (for all personal issues)
eeecs496-f24-staff@umich.edu
- [Piazza](#) for getting help
- [Canvas](#) for all important announcements to the class
- [Gradescope](#) or [Canvas](#) for submitting assignments

EECS 496: Major Design Experience Professionalism

“Design principles for multidisciplinary team projects, team strategies, entrepreneurial skills, ethics, social and environmental awareness, and life long learning.”

What it's for?

It's needed for ABET (Accreditation Board for Engineering and Technology)

This accreditation is the assurance that a university program meets the quality standards established for the engineering profession.

Many employers, graduate schools, and licensure, certification, and registration boards require graduation from an accredited program as a minimum qualification.

What it's for?

It's needed for ABET (Accreditation Board for Engineering and Technology)

Help you be successful in your career

- **Non-technical knowledge and skills**
- **What isn't taught in your other EECS courses**

What it's for?

It's needed for ABET (Accreditation Board for Engineering and Technology)

Help you be successful in your career

- Non-technical knowledge and skills
- What you didn't learn in all those other EECS courses

Becoming a professional

I invite subject experts from across campus and industry to explain and share their expertise with us

Being a Professional

- **What does it mean?**

[[Being A Professional: Dale Atkins at TEDxYouth@EHS](#)]

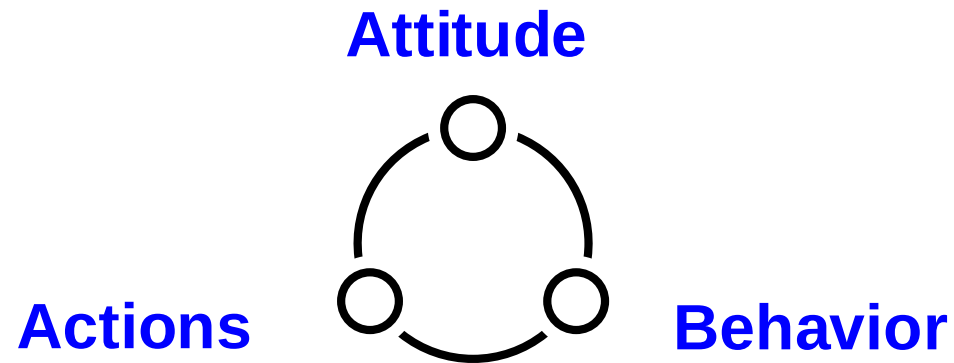


Being a Professional

- What differentiates being a Professional from e.g. a Technician?

Professionals deliver judgement (a considered decision)

- Professional is not just a title, it can also describe or mean:



Recent Examples

- **Aug 2023: Campus-wide shutdown of internet service**
 - First detected suspicious activity on Aug. 23,
 - Information Assurance team shut down the system Aug. 27
 - Limited service restored Aug. 30
- **July 2024: Crowdstrike**
 - Crowdstrike releases update to millions of Windows machines on July 19
 - Bug caused it to crash on boot, also crashing OS
 - \$B's of economic impact

Topics We'll Cover in 496

Working Life

Personal Finance and Investments

Business and Project Management

Intellectual Property and Patents

Ethics

Diversity, Equity, and Inclusion

Venture Capital

Entrepreneurship

Lecture Attendance

- **You are required to attend lecture in person**
 - This is a courtesy to our guest speakers
 - This is also required for assessment
- **Minimize disruptions when speakers are presenting**
 - No eating, drinking, being disruptive
 - Laptops or phones out during presentations is disrespectful / disruptive
- **In-person attendance will be part of your grade (10% total)**
- **In-class quizzes will be given during lecture (40% total)**
 - Will be based on lecture content that day
 - Bring a device that allows you to login to Canvas and take the quiz during class
 - Lowest 2 scores will be dropped

Absences

- If you miss a lecture, watch the recording (on Canvas)
- You are allowed 2 unexcused absences with no penalty to your attendance grade
 - The lowest 2 in-class quiz and attendance grades will be dropped
- Should you miss a 3rd lecture, you can submit a 1-page reflection on that lecture within 2 weeks of the missed lecture (this grade substitutes for the in-class quiz)
 - This option can only be used once
- Additional absences or missed quizzes will result in 0 points for those days

Homeworks

- **5 homeworks, 50 pts each (50% of grade)**
- **Late homework will be penalized:**
 - Homework will lose 15 points for each day late (24 hours)
 - Homework that is more than 48 hours late will be assigned a 0

Homework	Issued (Due)	Points
HW1 – Elevator Pitch	Sept 6 (Sept 20)	50
HW2 – Budgets	Sept 20 (Oct 4)	50
HW3 – Executive Summary	Oct 4 (Oct 18)	50
HW4 – Microaggressions	Oct 25 (Nov 8)	50
HW5 – Ethics Op-Ed	Nov 8 (Nov 22)	50
	Total	250

Generative AI Policy

New for Fall 2024...

- You are encouraged to use generative AI tools, such as <https://umgpt.umich.edu/>
- Assist with brainstorming ideas, conducting initial research, and refining language and grammar for homework assignments.
- All submitted work must represent your own understanding and original thought.
- If you use generative AI, you must include a brief statement in your assignment stating that generative AI was used.
- Using generative AI tools will not impact your grade, provided that their usage is properly disclosed and adheres to the guidelines.

Grading

In-class Quiz or One-Time Reflection, 20pts each (10 of 12)	200 pts	40%
In-class attendance, 5 pts each (10 of 12)	50 pts	10%
Homework, 50pts each (5)	250 pts	50%
Filling out teaching evaluation (extra credit)	5 pts	1%
Total	500 pts	

No midterm or final exams

Total Points	Grade
495-500	A+
466-494	A
450-465	A-
433-449	B+
416-432	B
400-415	B-
0-399	C+ or lower

Starting a Career

Working in different environments

- Academia – teaching, research, funding and service
- Government – research, advising, contracting
- Big Industry – teams, hierarchical structure, yearly reporting and evaluation, bottom line driven: profit and loss
- Small firms/startups – each individual wears many hats, lots of fire-fighting, value driven: demonstrating traction

In all of these career roles: respect, caring, creativity, leadership, teamwork, diligence, timeliness, curiosity and learning are desired.

Which is the right career for me?

What is important to you?

- Experience & Development
 - **Salary & Benefits**
 - Geographic Location
-
- Company Culture
 - Advancement
 - Research Opportunities

Salary & Benefits

Total Compensation: Entire benefit offerings a company provides an employee in exchange for work – this includes benefits (their value) and base salary.

- Base Salary
 - Annual Base Wage
- Bonus Eligibility
 - Performance Based (personal/division/company)
- Commissions / Incentives
- Sign-On Bonus
 - Offered at the time of hire
 - Taxable
- Stock Options – Need to vest
- Relocation – Taxable (or perhaps remote)

Salary & Benefits

Choice of Health Plans

Disability/Life Insurance

Time Off – Holiday, Vacation & Sick

Healthcare Flex Accounts

Tuition Reimbursement

Wellness Options – Gym, etc.

Other Perks: Product Discounts, Financial/Legal Services, Cell Phone, Laptop, Flexible Work Arrangements, Remote Working

Salary & Benefits

Is the salary offered in line with similar positions?

How do the benefits impact me now or in the future?

Find and use resources (LinkedIn, Glassdoor, Payscale, ECRC, colleagues, ...) to answer those questions

Research the Market

Electrical Engineer Salaries in Ann Arbor, MI ⓘ

Confident · 6,819 Salaries Submitted · Updated Jun 6, 2024

Experience

All years of experience

Industries

All industries

Total pay range

\$95K - \$143K/yr

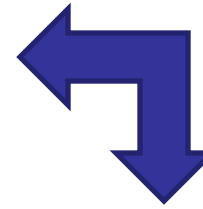
\$116K/yr Median total pay

Pay breakdown

\$78K - \$111K/yr Base pay

\$17K - \$33K/yr Additional pay

Why such a big difference?



- Geo differential
- Supply and demand
- Cost of living

Electrical Engineer Salaries in San Jose, CA ⓘ

Confident · 6,819 Salaries Submitted · Updated Jun 6, 2024

Experience

All years of experience

Industries

All industries

Total pay range

\$129K - \$201K/yr

\$160K/yr Median total pay

Pay breakdown

\$100K - \$147K/yr Base pay

\$29K - \$54K/yr Additional pay

[credit Glassdoor]

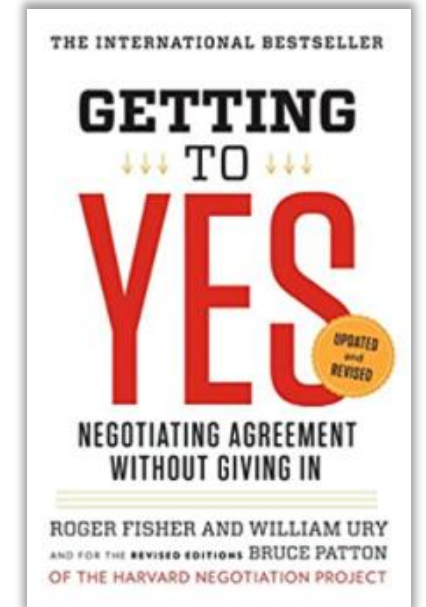
Negotiation

What you could negotiate (Base Salary, Start Date, Time Off, Professional Development, ...)

What you can use to negotiate (Experience, Education, Salary Data, Cost of Living, other job offers, ...)

Be careful:

- Don't be unrealistic with the requests – be data driven
- Address all questions and concerns during same conversation
- Be particularly careful with emails when you are negotiating (tone)
- Prefer phone calls or in person, but remember the confirmation email!



5 Keys to Negotiation

Neal Hartman: The five keys to negotiating your salary



Negotiation

Key is Preparation

- 1. Think about timing**
- 2. Do your homework. Know what's going on in the market**
- 3. Highlight your accomplishments (but not bragging)**
- 4. Don't compare your salary to your colleagues**
- 5. Go in with a positive attitude, and stay calm, especially if the outcome isn't what you want**

Job offer considerations & tips

- **When you receive an offer...**
 - Carefully evaluate all the aspects important for you
 - Understand what the deadline is for responding to the offer (in case you need more time, determine how much time you will need)
 - Ask questions

- **Always remember...**
 - “Get it in writing”
 - Find and use the right resources
 - Be respectful

Declining an offer

- Don't lose time. As soon as you've made a decision, let them know
- Prefer a phone call to email
- Decline an offer graciously (ask experts)
- Provide a reason, but don't get specific... "Less is more"
- Keep it simple! Be courteous and respectful!
- Leave the door open for the future

"Thank you so much for the offer and your time ... But I decided to pursue another opportunity at this time"

Which is the right career for me?

What is important to you?

- Experience & Development
- Salary & Benefits
- Geographic Location

- **Company Culture**
- Advancement
- Research Opportunities

Company Culture

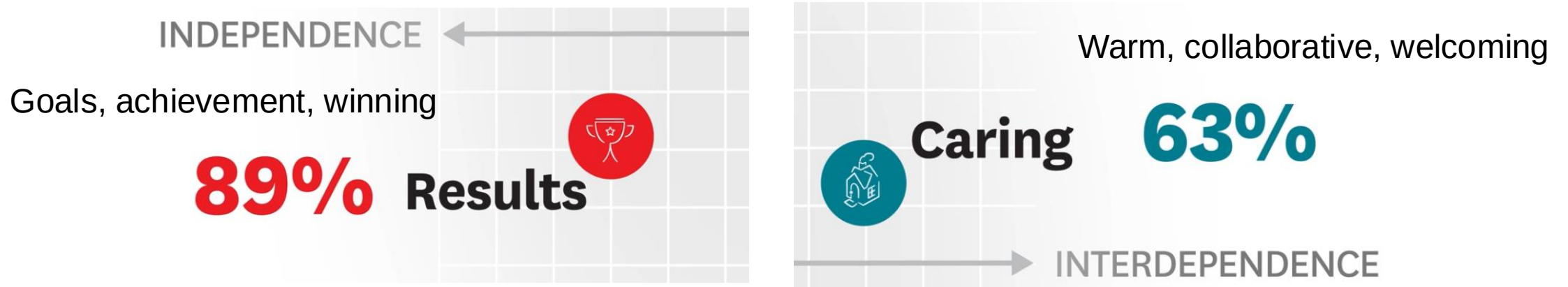
The 8 Types of Company Culture - Harvard Business Review

THE FRAMEWORK



Company Culture

- “Culture eats strategy for breakfast”
 - No matter how strong your strategic plan is, its efficacy will be held back if your team doesn’t share the proper culture
 - The people implementing the plan make all the difference

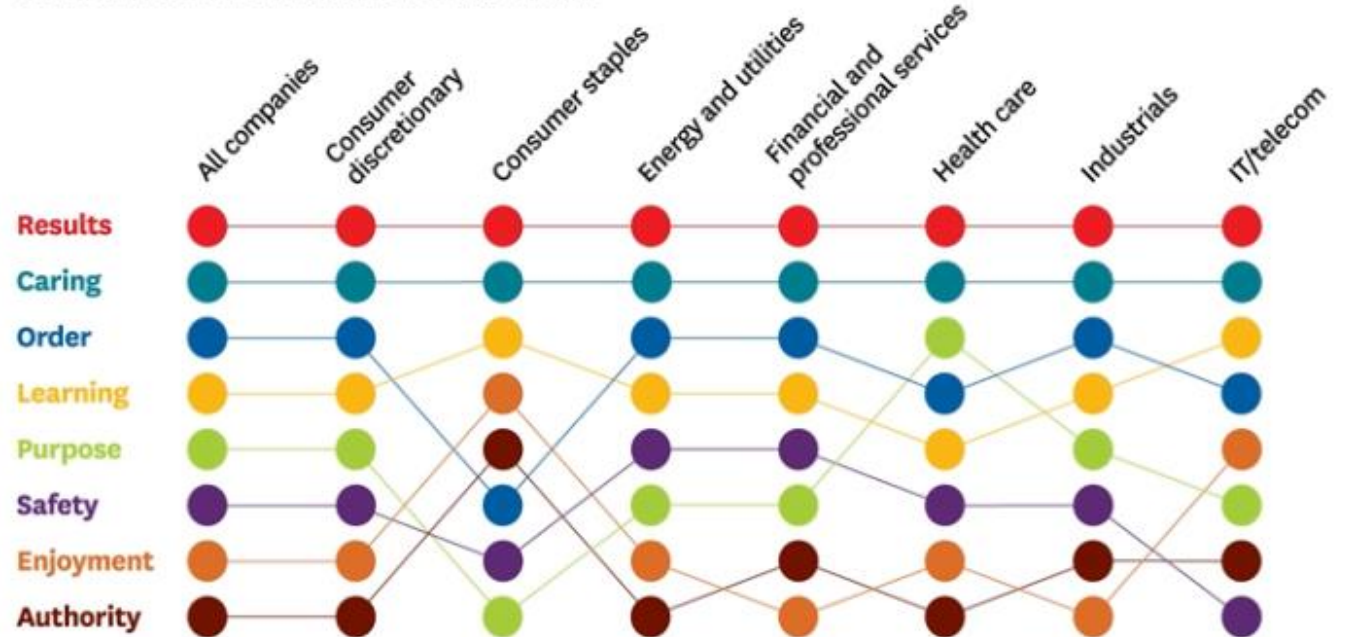


- **Results** and **Caring** consistently rank at or near the top of most people’s list

Company Culture – Final Thoughts

- Think about the type of company you want to work in
- Do your research
- Company culture is often not easily discovered
 - You have to ask people who work there

CULTURE STYLES RANKED BY INDUSTRY



Next Lecture: Interview Essentials and Evaluating Job Offers

